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Remark: Please record how much time (in minutes) you spent on each question. This information is important to estimate the workload of this assignment and the difficulty of each question. It will NOT affect your grade.

Question	1	2(a)	2(b)	3(a)	3(b)	3(c)	3(d)	4(a)	4(b)	5
Time(min)										

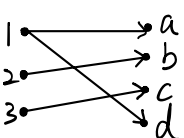
1. This question tests your understanding on set, relation, and function.

Let $A = \{1, 2, 3\}$ and $B = \{a, b, c, d\}$.

- (a) Determine the Cartesian product of A and B. (4 marks)
 (b) Define a subset $C \subseteq A \times B$ such that C contains exactly three elements. (2 marks)
 (c) Consider the relation R defined as $R = \{(1, a), (2, b), (3, c), (1, d)\}$. Represent R as a directed graph. Determine if R is a function from A to B. Justify your answer. (4 marks)

Ans: (a) $A \times B = \{(1, a), (1, b), (1, c), (1, d), (2, a), (2, b), (2, c), (2, d), (3, a), (3, b), (3, c), (3, d)\}$

(b) $C = \{(1, a), (1, b), (1, c)\}$

(c)  R is NOT a function from A to B, because $(1, a) \in R$ and $(1, d) \in R$ and $a \neq d$

2. This question tests your skills in using truth table. (10 marks: 5 marks each)

Please use truth tables to show that

- (a) $(p \rightarrow q) \rightarrow r \not\equiv p \rightarrow (\sim q \vee r)$
 (b) $p \rightarrow (q \rightarrow r) \equiv q \rightarrow (p \rightarrow r)$.

(a)

Ans:

P	q	r	$p \rightarrow q$	$(p \rightarrow q) \rightarrow r$	$\sim q$	$\sim q \vee r$	$p \rightarrow (\sim q \vee r)$
T	T	T	T	T	F	T	T
T	T	F	T	F	F	F	F
T	F	T	F	T	T	T	T
T	F	F	F	T	T	T	T
F	T	T	T	T	F	T	T
F	T	F	T	F	F	F	T
F	F	T	T	T	T	T	T
F	F	F	T	F	T	T	T

$(p \rightarrow q) \rightarrow r$ and $p \rightarrow (\sim q \vee r)$ have different truth values in the 6th and 8th rows, so $(p \rightarrow q) \rightarrow r \not\equiv p \rightarrow (\sim q \vee r)$

P	q	r	$q \rightarrow r$	$p \rightarrow (q \rightarrow r)$	$p \rightarrow r$	$q \rightarrow (p \rightarrow r)$
T	T	T	T	T	T	T
T	T	F	F	F	F	F
T	F	T	T	T	T	T
T	F	F	T	T	F	T
F	T	T	T	T	T	T
F	T	F	F	T	T	T
F	F	T	T	T	T	T
F	F	F	T	T	T	T

$p \rightarrow (q \rightarrow r)$ and $q \rightarrow (p \rightarrow r)$ always have the same truth values, so $p \rightarrow (q \rightarrow r) \equiv q \rightarrow (p \rightarrow r)$

3. Please simplify the below statements as much as you can. A simpler statement means that it uses a smaller number of logical connectives. The result should only use logical connectives from $\{\sim, \wedge, \vee\}$. Please show the detailed steps (You may refer to *Example 2.1.14* on page 50 of the textbook.). (20 marks: 5 marks each)

- (a) $((p \wedge q) \vee r) \vee (\sim p \vee (p \wedge \sim q))$ (a) $((p \wedge q) \vee r) \vee (\sim p \vee (p \wedge \sim q)) \equiv ((p \wedge q) \vee r) \vee ((\sim p \vee p) \wedge (\sim p \vee \sim q))$ by Distributive law
 (b) $(p \rightarrow (q \rightarrow r)) \rightarrow ((p \rightarrow q) \rightarrow (p \rightarrow r))$ $\equiv ((p \wedge q) \vee r) \vee (t \wedge (\sim p \vee \sim q))$ by Negation law
 (c) $((p \vee (q \rightarrow r)) \rightarrow p) \vee (q \wedge r)$ $\equiv ((p \wedge q) \vee r) \vee (\sim p \vee \sim q)$ by Identity law
 (d) $(p \wedge q) \rightarrow ((q \rightarrow r) \leftrightarrow \sim p)$ $\equiv ((p \wedge q) \vee r) \vee \sim(p \wedge q)$ by De Morgan's laws
 $\equiv ((p \wedge q) \vee \sim(p \wedge q)) \vee r$ by Associative/Commutative law
 $\equiv t \vee r$ by Negation law
 $\equiv t$ by Universal bound law

(b) Ans:

$$\begin{aligned}
 (p \rightarrow (q \rightarrow r)) \rightarrow ((p \rightarrow q) \rightarrow (p \rightarrow r)) &\equiv (p \rightarrow (\sim q \vee r)) \rightarrow ((\sim p \vee q) \rightarrow (\sim p \vee r)) \\
 &\equiv (\sim p \vee (\sim q \vee r)) \rightarrow (\sim(\sim p \vee q) \vee (\sim p \vee r)) \\
 &\equiv (\sim p \vee \sim q \vee r) \rightarrow ((p \wedge \sim q) \vee \sim p \vee r) \text{ by De Morgan's laws} \\
 &\equiv (\sim p \vee \sim q \vee r) \rightarrow (((p \wedge \sim q) \vee \sim p) \vee r) \text{ by Associative law} \\
 &\equiv (\sim p \vee \sim q \vee r) \rightarrow (((p \vee \sim p) \wedge (\sim q \vee \sim p)) \vee r) \text{ by Distributive law} \\
 &\equiv (\sim p \vee \sim q \vee r) \rightarrow ((t \wedge (\sim q \vee \sim p)) \vee r) \text{ by Negation law} \\
 &\equiv (\sim p \vee \sim q \vee r) \rightarrow (\sim p \vee \sim q \vee r) \text{ by Identity/Commutative law} \\
 &\equiv \sim(\sim p \vee \sim q \vee r) \vee (\sim p \vee \sim q \vee r) \\
 &\equiv t \text{ by Negation law}
 \end{aligned}$$

(c) $((p \vee (q \rightarrow r)) \rightarrow p) \vee (q \wedge r) \equiv ((p \vee (\sim q \vee r)) \rightarrow p) \vee (q \wedge r)$
 $\equiv (\sim(p \vee (\sim q \vee r)) \vee p) \vee (q \wedge r)$
 $\equiv ((\sim p \wedge (\sim q \wedge \sim r)) \vee p) \vee (q \wedge r) \text{ by De Morgan's laws}$
 $\equiv ((\sim p \vee p) \wedge ((\sim q \wedge \sim r) \vee p)) \vee (q \wedge r) \text{ by Distributive law}$
 $\equiv (t \wedge (p \vee (\sim q \wedge \sim r))) \vee (q \wedge r) \text{ by Negation/Commutative law}$
 $\equiv p \vee (\sim q \wedge \sim r) \vee (q \wedge r) \text{ by Identity law}$
 $\equiv p \vee (q \wedge (\sim r \vee r)) \text{ by Distributive law}$
 $\equiv p \vee (q \wedge t) \text{ by Negation law}$
 $\equiv p \vee q \text{ by Identity law}$

(d) $(p \wedge q) \rightarrow ((q \rightarrow r) \leftrightarrow \sim p) \equiv \sim(p \wedge q) \vee ((\sim q \vee r) \vee \sim p) \wedge (\sim(\sim p) \vee (\sim q \vee r))$
 $\equiv (\sim p \vee \sim q) \vee ((q \wedge \sim r) \vee \sim p) \wedge (p \vee \sim q \vee r) \text{ by De Morgan's laws}$
 $\equiv ((\sim p \vee \sim q) \vee ((q \wedge \sim r) \vee \sim p)) \wedge ((\sim p \vee \sim q) \vee (p \vee r \vee q)) \text{ by Distributive/Commutative law}$
 $\equiv (\sim p \vee (\sim q \vee (q \wedge \sim r))) \wedge (\sim q \vee (\sim p \vee (p \vee r))) \text{ by Distributive law}$
 $\equiv (\sim p \vee ((\sim q \vee q) \wedge (\sim q \wedge \sim r))) \wedge (\sim q \vee (\sim p \vee p \vee r)) \text{ by Distributive/Associative law}$
 $\equiv (\sim p \vee (t \wedge (\sim q \wedge \sim r))) \wedge (\sim q \vee t \vee r) \text{ by Negation law}$

$$\equiv (\sim P \vee \sim Q \vee \sim R) \wedge (\sim Q \vee (t \vee r)) \quad \text{by Identity law}$$

$$\equiv (\sim P \vee \sim Q \vee \sim R) \wedge (\sim Q \vee t) \quad \text{by Universal bound law}$$

$$\equiv (\sim P \vee \sim Q \vee \sim R) \wedge t \quad \text{by Universal bound law}$$

$$\equiv \sim P \vee \sim Q \vee \sim R \quad \text{by Identity law}$$

4. This question tests your skills in deduction reasoning by rules of inference. You need to deduce the conclusion from the given premises using the valid argument forms listed in Table 2.3.1. Give a reason for each step as in Example 2.3.8. (20 marks: 10 marks each)

(a) Premises: $\sim p \vee q \rightarrow r$; $s \vee \sim q$; $\sim t$; $p \rightarrow t$; $\sim p \wedge r \rightarrow \sim s$;
Conclusion: $\sim q$

(b) Premises: $(p \rightarrow q) \wedge (r \rightarrow s)$; $(p \rightarrow w) \wedge (s \rightarrow x)$; $\sim(w \wedge x)$; r
Conclusion: $\sim p$

(a) Premises: a. $\sim p \vee q \rightarrow r$, b. $s \vee \sim q$, c. $\sim t$, d. $p \rightarrow t$, e. $\sim p \wedge r \rightarrow \sim s$

Ans: (1) $p \rightarrow t$ by premise (d)
 $\sim t$ by premise (c)
 $\therefore \sim p$ by modus tollens
 (2) $\sim p$ by (1)
 $\therefore \sim p \vee q$ by generalization
 (3) $\sim p \vee q \rightarrow r$ by premise (a)
 $\sim p \vee q$ by (2)
 $\therefore r$ by modus ponens
 (4) $\sim p$ by (1)
 r by (3)
 $\therefore \sim p \wedge r$ by conjunction
 (5) $\sim p \wedge r \rightarrow \sim s$ by premise (e)
 $\sim p \wedge r$ by (4)
 $\therefore \sim s$ by modus ponens
 (6) $s \vee \sim q$ by premise (b)
 $\sim s$ by (5)
 $\therefore \sim q$ by elimination

(b) Premises: a. $(p \rightarrow q) \wedge (r \rightarrow s)$ b. $(p \rightarrow w) \wedge (s \rightarrow x)$ c. $\sim(w \wedge x)$ d. r
 (1) $(p \rightarrow q) \wedge (r \rightarrow s)$ by premise (a)
 $\therefore r \rightarrow s$ by specialization
 (2) $r \rightarrow s$ by (1)
 r by premise (d)
 $\therefore s$ by modus ponens
 (3) $(p \rightarrow w) \wedge (s \rightarrow x)$ by premise (b)
 $\therefore s \rightarrow x$ by specialization
 (4) $s \rightarrow x$ by (3)
 s by (2)
 $\therefore x$ by modus ponens
 (5) $\sim(w \wedge x)$ by premise (c)
 $\therefore \sim w \vee \sim x$ by De Morgan's laws
 (6) $\sim w \vee \sim x$ by (5)
 x by (4)
 $\therefore \sim w$ by elimination
 (7) $(p \rightarrow w) \wedge (s \rightarrow x)$ by premise (b)
 $\therefore p \rightarrow w$ by specialization
 (8) $p \rightarrow w$ by (7)
 $\sim w$ by (6)
 $\therefore \sim p$ by modus tollens

5. Puzzle problem: Given the following information about a computer program, find the mistake in the program.

- There is an undeclared variable or there is a syntax error in the first five lines.
- If there is a syntax error in the first five lines, then there is a missing semicolon or a variable name is misspelled.
- There is not a missing semicolon.
- There is not a misspelled variable name.

Please justify your answer with the help of propositional logic. (10 marks)

Let u : There is an undeclared variable. s : There is a syntax error in the first five lines. m : There is a missing semicolon. p : A variable name is misspelled
Then premises can be translate to a) $u \vee s$ b) $s \rightarrow (m \vee p)$ c) $\sim m$ d) $\sim p$

(1) $\sim m$ by premise (c)

$\sim p$ by premise (d)

$\therefore \sim m \wedge \sim p$ by conjunction

(2) $\sim m \wedge \sim p$ by (1)

$\therefore \sim (m \vee p)$ by De Morgan's law

(3) $\sim (m \vee p)$ by (2)

$s \rightarrow (m \vee p)$ by premise (b)

$\therefore \sim s$ by modus tollens

(4) $u \vee s$ by premise (a)

$\sim s$ by (3)

$\therefore u$ by elimination

Therefore, the mistake in the program is: There is an undeclared variable.